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A nonlinear association and threshold effect of healthy eating index-2015 with hypertension in US children and adolescents: a cross-sectional analysis of NHANES 1999–2020

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Abstract

Background Pediatric hypertension is an increasingly recognized health concern with long-term cardiovascular implications. Diet quality is a modifiable factor that may influence blood pressure in youth. However, research on the association between diet quality and hypertension in Children and Adolescents remains limited. This study aims to assess the potential nonlinear association between diet quality, measured by the Healthy Eating Index 2015 (HEI-2015) score, and the potential threshold effect with hypertension in US Children and Adolescents.

Methods We conducted a cross-sectional study using data from the National Health and Nutrition Examination Survey (NHANES) 1999–2020. A total of 17,528 participants aged 8–18 years were included after excluding those with missing diet or blood pressure data and other ineligible cases. Diet quality was assessed by Healthy Eating Index-2015 (HEI-2015) scores (range 0–100). Multivariable logistic regression was used to evaluate the association between HEI-2015 and hypertension. We examined potential non-linear relationships using a two-piecewise linear spline approach to identify any threshold effect. Subgroup and sensitivity analyses were performed to assess the robustness of the findings.

Results The analysis included 17,528 children and adolescents aged 8–18 years from NHANES 1999–2020. The prevalence of hypertension was 6.14% ($n = 1,076$). The participants were classified according to HEI-2015 tertiles (T1: 12.36–41.47; T2: 41.47–51.28; T3: 51.28–95.65). Compared with T2 and T3 group, participants in the lowest tertile of the HEI-2015, had higher prevalence of hypertension (6.73% vs. 6.23% and 5.56%). Specifically, below an HEI-2015 score of 65.13, each 10-point increase in HEI-2015 was associated with a 7% lower prevalence of hypertension (adjusted OR = 0.93, 95% CI: 0.87–0.99; $p = 0.023$). In contrast, above this threshold (HEI-2015 ≥ 65.13), each 10-point increase was associated with a much higher (52%) prevalence of hypertension (OR = 0.48, 95% CI: 0.28–0.83; $p = 0.009$). This analysis reveals a nonlinear association between HEI-2015 and pediatric hypertension, with an inflection point

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identified at an HEI-2015 score of ≈ 65 . A formal test for threshold nonlinearity was significant ($p=0.012$), confirming a change in slope at an HEI-2015 score of ≈ 65 . Sensitivity analyses yielded consistent results.

Conclusions In this cross-sectional study, a higher HEI-2015 score was nonlinearly associated with a lower prevalence of hypertension in US children and adolescents, with an observed inflection point near a score of 65. This finding provides a specific, quantifiable hypothesis for future prospective studies investigating dietary thresholds for hypertension prevention in children and adolescents.

Contributions to the Literature

Evidence supporting specific, quantifiable dietary thresholds for hypertension prevention in children and adolescents remains limited. The study identifies a specific HEI-2015 score threshold (≈ 65) that maybe associated with a lower prevalence of hypertension in children and adolescents.

The study provides a data-derived target (HEI-2015 ≈ 65), which establishes a hypothesis for future longitudinal and intervention studies to validate its predictive value and explore its potential utility in preventing hypertension in children and adolescents.

Keywords Healthy Eating Index, Diet, Hypertension, Children, NHANES

Introduction

Hypertension in children and adolescents is a significant and growing public health concern [1, 2]. Once considered rare, its prevalence has risen alongside childhood obesity [3, 4]. According to updated American Academy of Pediatrics (AAP) criteria, about 1 in 7 adolescents (12–19 years old) now have elevated blood pressure or hypertension, with prevalence varying by weight status, 2% in healthy-weight youth versus 14% in severe obesity [5]. This condition is clinically concerning because childhood hypertension often persists into adulthood and is linked to early cardiovascular risk markers if untreated [6, 7]. In response, the 2017 AAP guidelines lowered diagnostic thresholds, and emphasized lifestyle intervention as a cornerstone of management, including recommendations for overall dietary improvement for at risk youth [8, 9]. Poor diet quality, characterized by excessive sodium, added sugars, saturated fats alongside insufficient intake of fruits, vegetables, whole grains, serve as a modifiable risk factor for prevalence of hypertension [10, 14].

The Healthy Eating Index-2015 (HEI-2015) provides a validated 100-point composite measure of diet quality [15, 16], evaluating both adequacy (fruits, vegetables, whole grains) and moderation components (sodium, added sugars, saturated fats). In adults, higher HEI-2015 scores are consistently associated with cardioprotective effects, including a lower risk of hypertension [10, 17, 21]. Emerging evidence, however, suggests similar benefits in youth, with healthier dietary patterns (HEI-2015/DASH-based) associated with significantly lower risks of elevated blood pressure and metabolic syndrome [5, 22, 23]. However, the existing pediatric evidence remains incomplete and occasionally inconsistent. Some cross-sectional studies support a beneficial association between diet quality and blood pressure in youth [24, 25], whereas other well-designed prospective cohort studies have not found a significant association between early diet quality

scores and later blood pressure [26]. A systematic review also noted that the strength of evidence linking dietary quality indices to blood pressure in adolescents is mixed [27].

However, inconsistencies in prior studies may arise from an oversimplified linear modeling approach. This approach fails to capture the true complexity of the relationship. Emerging evidence indicates that the association between diet quality and health outcomes may involve threshold effects rather than a simple linear pattern. For instance, in the general U.S. adult population, the relationship between the HEI-2015 and hypertension is significantly nonlinear and inverse. Notably, the pattern differs by gender, with a U-shaped curve observed among males [10]. More relevant to pediatric populations, a study of school-aged children found that an HEI-2015 score above the 33rd percentile was linked to a substantially reduced risk of insulin resistance, a key cardiometabolic risk factor [24]. These findings suggest that the health benefits of improved diet quality in youth may not follow a linear pattern, but instead increase in magnitude after crossing a particular quality threshold. The potential nonlinear association and threshold effect between diet quality and hypertension in children remains unclear.

Therefore, we examined the association between HEI-2015 scores and hypertension prevalence among children and adolescents using two decades of NHANES data (1999–2020). We aimed to investigate the potential nonlinear association and to identify a threshold for the HEI-2015 score in relation to hypertension among US children and adolescents.

Methods

Data source and study population

We performed a cross-sectional analysis of participants from the National Health and Nutrition Examination Survey (NHANES), spanning survey years 1999 through

2020. NHANES is a continuous series of stratified, multistage probability surveys designed to be representative of the civilian, non-institutionalized US population. The survey consists of interviews, physical examinations, and laboratory tests, and it is conducted in repeated two-year cycles by the National Center for Health Statistics. This analysis utilized publicly available, de-identified NHANES data and was deemed exempt from additional Institutional Review Board review. All participants (or their guardians) provided written informed consent for NHANES; the procedures were approved by the National Center for Health Statistics (NCHS) ethics review board. All of the NHANES data is accessible to the public and can be downloaded freely through:

<https://www.cdc.gov/nchs/nhanes/about/index.html>.

For this study, we aggregated data from all NHANES cycles between 1999 and 2020 that included the necessary variables. The study sample was restricted to children and adolescents aged >8 to <18 years. We set the lower age limit at 8 years because reliable dietary recall data in NHANES are available starting around age 8 (younger children's data often involve proxy reporting). We further excluded individuals with missing or incomplete data on dietary intake ($n=6444$), blood pressure measurements ($n=657$). After applying inclusion and exclusion criteria, the final analytic sample consisted of $N=17,528$ children and adolescents. The study flowchart was presented in Fig. 1.

Dietary assessment and HEI-2015 calculation

Detailed dietary intake data were collected from NHANES participants through two 24-h dietary recalls, which were conducted by a trained interviewer face-to-face in the Mobile Examination Center (MEC) on the first day and a follow-up interview 3–10 days later via phone including both weekdays and weekends, to capture intra-individual variation in dietary intake. More details on dietary surveys and quality control can be found in the survey manual. <https://wwwn.cdc.gov/nchs/nhanes/analyticguidelines.aspx>. These two sets of dietary information were synthesized to assess overall diet quality and calculate multiple dietary quality indicators, including HEI-2015 [24, 28]. The HEI-2015 total and component scores were calculated by the simple HEI scoring algorithm using publicly available SAS macros [29]. The HEI-2015 comprises 13 dietary components: 9 adequacy components (Total Fruits, Whole Fruits, Total Vegetables, Greens and Beans, Whole Grains, Dairy, Total Protein Foods, Seafood/Plant Proteins, and Fatty Acids ratio) and 4 moderation components (Refined Grains, Sodium, Added Sugars, and Saturated Fats). Each component is scored on a scale (maximum 5 or 10 points) proportional to the amount consumed, with higher intake of adequacy components and lower intake of moderation components yielding higher scores. Component scores are summed to produce a total HEI-2015 score ranging from 0 to 100, with 100 indicating perfect adherence to

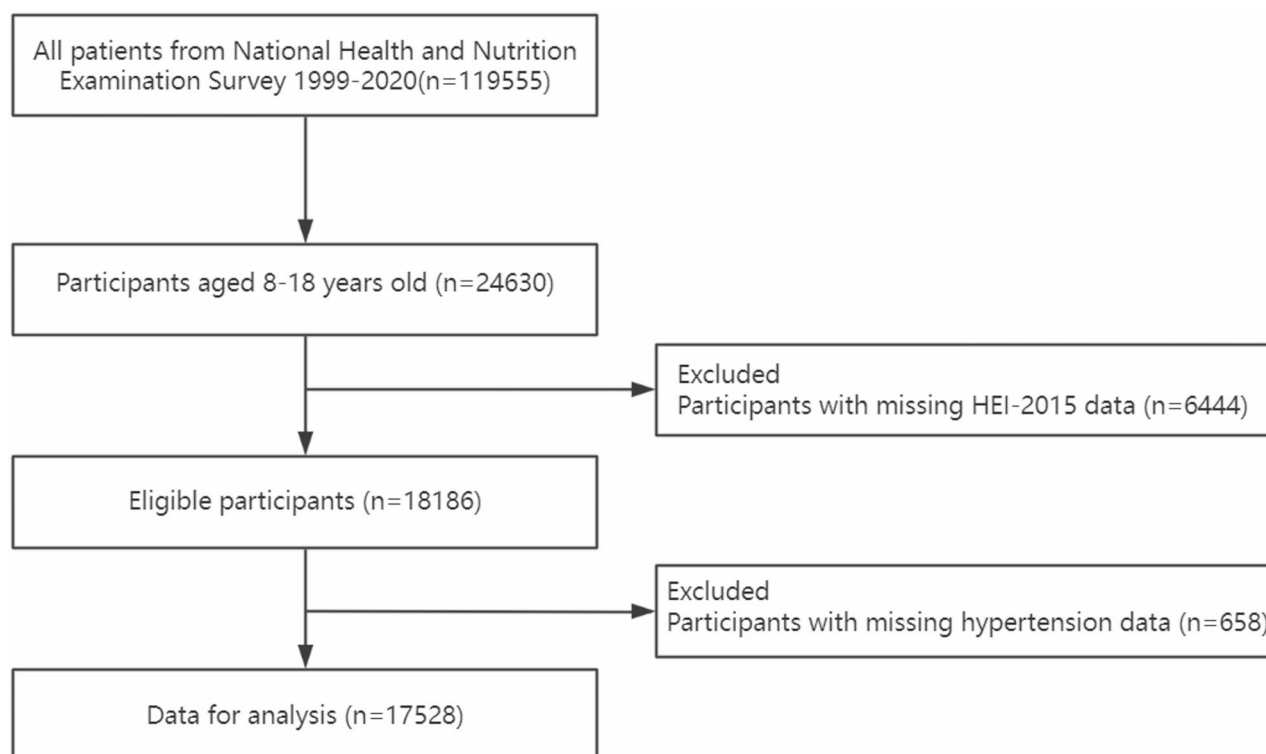


Fig. 1 Flowchart of participants selection. HEI-2015, Healthy Eating Index 2015

the Dietary Guidelines (a very healthy diet) and 0 indicating a very poor diet [16, 18, 30]. The use of HEI-2015 as a composite exposure obscures the contribution of individual dietary components, limiting interpretability of the observed associations. For example, a high score reflects high consumption of fruits, vegetables, and whole grains, adequate protein and dairy, appropriate unsaturated-to-saturated fat ratio, and low consumption of added sugar, salt, and saturated fat.

Blood pressure measurement and hypertension definition

Blood pressure (BP) was measured by trained clinicians during the NHANES physical examination following standard AHA protocols. Participants rested quietly seated for 5 min, and then three brachial blood pressure readings were obtained on the right arm (unless contraindicated) using an appropriate pediatric cuff size and a mercury sphygmomanometer or calibrated oscillometric device. The average of the three readings was used as the individual's systolic blood pressure (SBP) and diastolic blood pressure (DBP) for classification. If fewer than three readings were recorded (e.g. due to measurement difficulty), the available readings were averaged. We defined hypertension in children and adolescents according to the 2017 AAP Clinical Practice Guideline criteria [31] (which updated the Fourth Report from 2004). Specifically, for participants aged < 13 years, hypertension was defined as an average SBP or DBP at or above the 95th percentile for their age, sex, and height percentile, based on the updated reference tables derived from normal-weight children. Height percentiles for participants under 13 years were determined based on the CDC 2000 Growth Charts, per the 2017 AAP guideline. For adolescents 13–18 years old, hypertension was defined according to the 2017 AAP clinical practice guideline as SBP $\geq$ 130 mmHg or DBP $\geq$ 80 mmHg (equivalent to the 95th percentile for 13 years old children) or higher, which uses the same blood pressure thresholds ($\geq$ 130/80 mmHg for stage 1 hypertension) as the American College of Cardiology/American Heart Association (ACC/AHA) guidelines for adults [32]. In addition, any youth reporting a previous physician diagnosis of hypertension or current use of antihypertensive medication was classified as having hypertension.

Covariates

Covariates were selected based on prior literature indicating their association with both diet quality and blood pressure [33, 34]. These covariates (all assessed via NHANES questionnaires or examination) were: sex, age, race/ethnicity, poverty income ratio, education level, Body Mass Index (BMI) Z-score, total physical activity, etc. Given its strong association with both diet and blood pressure, BMI Z-score was included in the model to

account for this clinical correlation, even though it might also be a mediator.

Race/ethnicity were divided into five categories: non-Hispanic White, non-Hispanic Black, Mexican American, Other Hispanic, or other races. The following educational levels were included: below high school, high school, and above high school. Socioeconomic status was approximated by the poverty income ratio (PIR), a ratio of household income to the federal poverty level (adjusted for family size). PIR were divided into: Poor, Nearly poor, Middle income, High income. Body mass index (BMI) was calculated as weight (kg) divided by height (m) squared, using measured weight and height from the exam. For age < 20, BMI was interpreted via percentiles, we included BMI Z-score or percentile equivalently to adjust for body size.

Statistical analysis

This study utilized cross-sectional data from NHANES. It is important to note that the findings represent observational associations and cannot establish causality or determine the direction of the relationships observed. Means and standard deviations were calculated for continuous variables and proportions for categorical variables. Dummy variables were used to indicate missing covariate values. Baseline characteristics were compared across HEI-2015 tertiles using Kruskal-Wallis *H* test for continuous variables and the chi-square test (or Fisher's exact test) for categorical variables. A logistic regression model was performed to estimate the association between the HEI-2015 and hypertension. The primary exposure, HEI-2015 score, was modeled both as a continuous variable and in categories (tertiles) for descriptive purposes. In the main models, we treated HEI-2015 as continuous and report the odds ratio (OR) for hypertension per a 10-point increment in HEI score, which roughly corresponds to a difference that might be achieved by making moderate dietary improvements. Considering the clinical significance, we adjusted for the following covariates: age, sex, race/ethnicity, poverty income ratio, education level, and BMI Z-score.

A generalized additive model (GAM) was used to explore potential non-linear association between the HEI-2015 and hypertension. The spline analysis suggested a change in slope around the mid-range of HEI scores. We then used a two-piecewise linear regression model to examine the threshold effect of the HEI-2015 on hypertension [35]. The turning point for the HEI-2015 was determined using "exploratory" analyses, which is to move the trial turning point along the pre-defined interval and pick up the one which gave maximum model likelihood. We also performed a log-likelihood ratio test and compared the one-line linear regression model with the two-piecewise linear model. The bootstrap resampling

method was used to calculate the 95% CI for the turning point. Subgroup analyses in terms of age, sex, race/ethnicity, BMI Z-score, etc. were conducted to verify whether the association between HEI-2015 and hypertension remained stable across different subgroups. A sensitivity analysis was performed by applying pre-specified HEI-2015 cut-offs [30] in the piecewise regression model to test the stability of the observed threshold effect independent of data-driven selection. To address the concern that adjusting for body mass index (BMI) Z-score might introduce overadjustment bias (as BMI could be a mediator on the causal pathway), we performed a sensitivity analysis by refitting the primary multivariable logistic regression models without including BMI Z-score as a covariate. Survey-weighted analyses were performed to account for the NHANES complex survey design, including the baseline characteristics and threshold effect analysis. All the statistical analyses were performed using the EmpowerStats (www.empowerstats.com, X&Y solutions, Inc. Boston MA) and R software version 4.2.0 (<http://www.r-project.org>) and a two-tailed P -value < 0.05 was considered statistically significant.

Results

Participant characteristics

The baseline characteristics of the research participants was shown in Table 1. The analysis included 17,528 children and adolescents aged 8 to 18 years from NHANES 1999–2020. 8134 participants (46.41%) under the age of 13, and 9394 participants (53.59%) over the age of 13. There were 8762 (49.99%) female participants. The racial/ethnic composition was 27.49% non-Hispanic White, 28.62% non-Hispanic Black, 35.34% Hispanic (inclusive of Mexican American and other Hispanic), and 8.56% other races, reflecting the diversity of the US population of youth. This group had a prevalence of 6.14% of hypertension. We classified the participants according to their HEI-2015 tertiles (T1:12.36–41.47; T2:41.47–51.28; T3:51.28–95.65.). There is a significant difference among T1, T2, and T3 group in age, sex, family PIR, education level, BMI Z-score, total physical activity, etc. ($P < 0.05$). Compared with T2 and T3 group, participants in the lowest tertile of the HEI-2015, had higher prevalence of hypertension (6.73% vs. 6.23% and 5.56%). Survey-weighted analyses were performed to account for the NHANES complex survey design, and the core findings remained consistent, Supplementary Table 1.

Association between HEI-2015 and hypertension

Table 2 shows the results of the multivariate logistic regression models. When modeling HEI-2015 as a continuous variable across its full range, each 10-point increase in score was associated with a

10% lower odds of hypertension (adjusted OR = 0.90, 95% CI: 0.85–0.95; $P = 0.0002$) after adjusting for sex, age, race/ethnicity, PIR, parental education, and BMI Z-score. The P for trend across the model was < 0.05 . Given the potential non-linear relationship, we examined the association by categorizing participants into HEI-2015 tertiles. Compared to the low HEI-2015 group (12.36–41.47), the high and middle HEI-2015 group (range 41.47–51.28 and 51.28–95.65) had a lower odds of hypertension (OR = 0.83, 95% CI 0.71–0.97; OR = 0.94, 95% CI 0.81–1.09).

A sensitivity analysis was conducted to assess the possibility that BMI Z-score, a potential mediator, could introduce overadjustment bias and attenuate the true association. The association between HEI-2015 and hypertension was robust in the sensitivity analysis that excluded BMI Z-score from the model (Supplementary Table 2). The effect estimates remained virtually unchanged: the OR for hypertension per 10-unit increase in HEI-2015 was 0.90 (95% CI: 0.85, 0.95) in both the primary and the sensitivity model.

Identification of nonlinear relationship

We observed a nonlinear relationship between the HEI-2015 and hypertension (Fig. 2; Table 3). Using two-piecewise linear logistic models, we identified a threshold HEI-2015 score of approximately 65.1 (95% CI: 60.15–66.87) as the point at which the association with hypertension changed. When we allowed for different slopes below and above HEI = 65.13, the model fit improved significantly compared to a single linear term. The Logarithm Likelihood Ratio Test (LRT) corroborated the non-linear model's superiority with a significant P -value of 0.012, affirming the relevance of threshold effects in dietary assessments. Sensitivity analysis was performed excluding the adjustment for BMI Z-score, and the result was consistent (Supplementary Fig. 1). Survey-weighted analyses were performed and the core findings remained consistent, Supplementary Table 3.

Below the threshold (HEI-2015 < 65.13): The association between diet quality and hypertension was weakly inverse. Each 10-point increase in HEI-2015 score was associated with a 7% reduction in the odds of hypertension (adjusted OR = 0.93, 95% CI: 0.87–0.99).

Above the threshold (HEI-2015 ≥ 65.13): The inverse relationship between diet quality and hypertension became much stronger. For each 10-point increase in HEI above 65, the adjusted OR for hypertension was 0.48 (95% CI: 0.28–0.83; $p = 0.0086$). This suggests that among children already consuming a relatively healthy diet (HEI-2015 ≥ 65.13), further improvements in diet quality have a large payoff in terms of blood pressure reduction.

Table 1 Characteristics of the study population according to the tertiles of the HEI-2015 from NHANES 1999–2020 ($n = 17528$)

Parameters	HEI-2015				P value
	Total	Tertile 1	Tertile 2	Tertile 3	
	$n = 17,528$	$n = 5843(33.33\%)$	$n = 5842(33.33\%)$	$n = 5843(33.33\%)$	
Age (years)		12.36–41.47	41.47–51.28	51.28–95.65	< 0.001
< 13	8134 (46.41%)	2424 (41.49%)	2705 (46.30%)	3005 (51.43%)	
≥ 13	9394 (53.59%)	3419 (58.51%)	3137 (53.70%)	2838 (48.57%)	
Sex					< 0.001
Male	8766 (50.01%)	3062 (52.40%)	2963 (50.72%)	2741 (46.91%)	
Female	8762 (49.99%)	2781 (47.60%)	2879 (49.28%)	3102 (53.09%)	
Ethnicity					< 0.001
Non-Hispanic White	4818 (27.49%)	1815 (31.06%)	1594 (27.29%)	1409 (24.11%)	
Non-Hispanic Black	5016 (28.62%)	1860 (31.83%)	1760 (30.13%)	1396 (23.89%)	
Mexican American	4893 (27.92%)	1395 (23.87%)	1630 (27.90%)	1868 (31.97%)	
Other Hispanic	1300 (7.42%)	362 (6.20%)	378 (6.47%)	560 (9.58%)	
Other Race	1501 (8.56%)	411 (7.03%)	480 (8.22%)	610 (10.44%)	
Poverty income ratio					< 0.001
Poor	4926 (28.10%)	1593 (27.26%)	1691 (28.95%)	1642 (28.10%)	
Nearly poor	4440 (25.33%)	1581 (27.06%)	1448 (24.79%)	1411 (24.15%)	
Middle income	4101 (23.40%)	1399 (23.94%)	1407 (24.08%)	1295 (22.16%)	
High income	2802 (15.99%)	871 (14.91%)	885 (15.15%)	1046 (17.90%)	
Missing	1259 (7.18%)	399 (6.83%)	411 (7.04%)	449 (7.68%)	
EDU					< 0.001
Below high school	11,408 (65.08%)	3673 (62.86%)	3821 (65.41%)	3914 (66.99%)	
High school	4296 (24.51%)	1576 (26.97%)	1430 (24.48%)	1290 (22.08%)	
Above high school	18 (0.10%)	7 (0.12%)	7 (0.12%)	4 (0.07%)	
Missing	1806 (10.30%)	587 (10.05%)	584 (10.00%)	635 (10.87%)	
Total physical activity (MET/week)					< 0.001
< 600	4364 (24.90%)	1660 (28.41%)	1455 (24.91%)	1249 (21.38%)	
≥ 600	4191 (23.91%)	1377 (23.57%)	1412 (24.17%)	1402 (23.99%)	
Missing	8973 (51.19%)	2806 (48.02%)	2975 (50.92%)	3192 (54.63%)	
Height (cm)	155.87 ± 14.82	157.35 ± 14.54	156.00 ± 15.03	154.25 ± 14.71	< 0.001
Weight (kg)	55.49 ± 20.73	57.05 ± 20.99	55.53 ± 20.60	53.88 ± 20.48	< 0.001
BMI (kg/m ²)Z-score	-0.00 ± 1.00	0.04 ± 1.02	-0.00 ± 0.98	-0.04 ± 1.00	< 0.001
WBC	6.94 ± 2.06	6.87 ± 2.03	6.91 ± 2.05	7.02 ± 2.10	< 0.001
LYMPHOCYTE	2.38 ± 0.70	2.34 ± 0.68	2.38 ± 0.70	2.43 ± 0.72	< 0.001
SBP	105.95 ± 10.32	106.65 ± 10.43	106.21 ± 10.39	104.99 ± 10.07	< 0.001
DBP	58.45 ± 11.27	59.22 ± 11.32	58.34 ± 11.21	57.78 ± 11.25	< 0.001
Hypertension					0.016
No	16,452(93.86%)	5450 (93.27%)	5478 (93.77%)	5524 (94.54%)	
Yes	1076(6.14%)	393 (6.73%)	364 (6.23%)	319 (5.46%)	

Data were expressed as the mean ± SD, median (interquartile range), or percentage. Among the 17,528 individuals, the amount of missing values for the covariates were 1259 (7.18%) for Poverty income ratio, 1806(10.3%) for education, 8973(51.19%) for total physical activity, 20(0.11%) for Weight, 20(0.11%) for BMI, 154 (0.88%) for SBP Systolic Blood Pressure, 1664(9.49%) for WBC, 1672 (9.54%) for LYMPHOCYTE

HEI, healthy eating index; EDU, education; BMI, body mass index; WBC, white blood cell; SBP, systolic blood pressure; DBP, diastolic blood pressure

The continuous variable *P*-value is derived from Kruskal-Wallis *H* test. Fisher Exact test for categorical variables

Sensitivity analysis We have conducted a sensitivity analysis using pre-specified HEI-2015 cut-offs (the quartiles and HEI categories used in previous work [30]). The results consistently showed a significant change in the strength of association around these fixed points. we have included the complete results of this analysis in Table 4.

Subgroup analysis

Subgroup analyses revealed that higher HEI-2015 scores were consistently associated with reduced odds of hypertension across most demographic groups, as shown in Fig. 3. Significant inverse associations were observed in both sexes (males: OR 0.90, $p = 0.005$; females: OR 0.89, $p = 0.006$), all age groups (< 13 years: OR 0.89, $p = 0.004$;

Table 2 Association of HEI-2015 with prevalence of hypertension in individuals from NHANES 1999–2020

HEI-2015	Number of hypertension cases/ Total participants(prevalence)	Model I		Model II	
		OR (95% CI)	P-Value	OR (95% CI)	P-Value
HEI-2015 per10 unit increase	1076/17,528 (6.14%)	0.90 (0.85, 0.95)	0.0002	0.90 (0.85, 0.95)	0.0002
HEI-2015 tertile					
Low	393/5843 (6.73%)	Ref		Ref	
Middle	364/5842 (6.23%)	0.93 (0.80, 1.08)	0.3517	0.94 (0.81, 1.09)	0.4187
High	319/5843 (5.46%)	0.83 (0.71, 0.96)	0.0152	0.83 (0.71, 0.97)	0.0180
P for trend		0.0156		0.0187	

Data were presented as OR [95% CI]

Model I was adjusted for age and sex. Model II was adjusted for age, sex, race/ethnicity, poverty income ratio, education level, and Body Mass Index(kg/m²) Z-score

HEI-2015, healthy eating index-2015; BMI, Body Mass Index; OR, odds ratio; CI, confidence interval

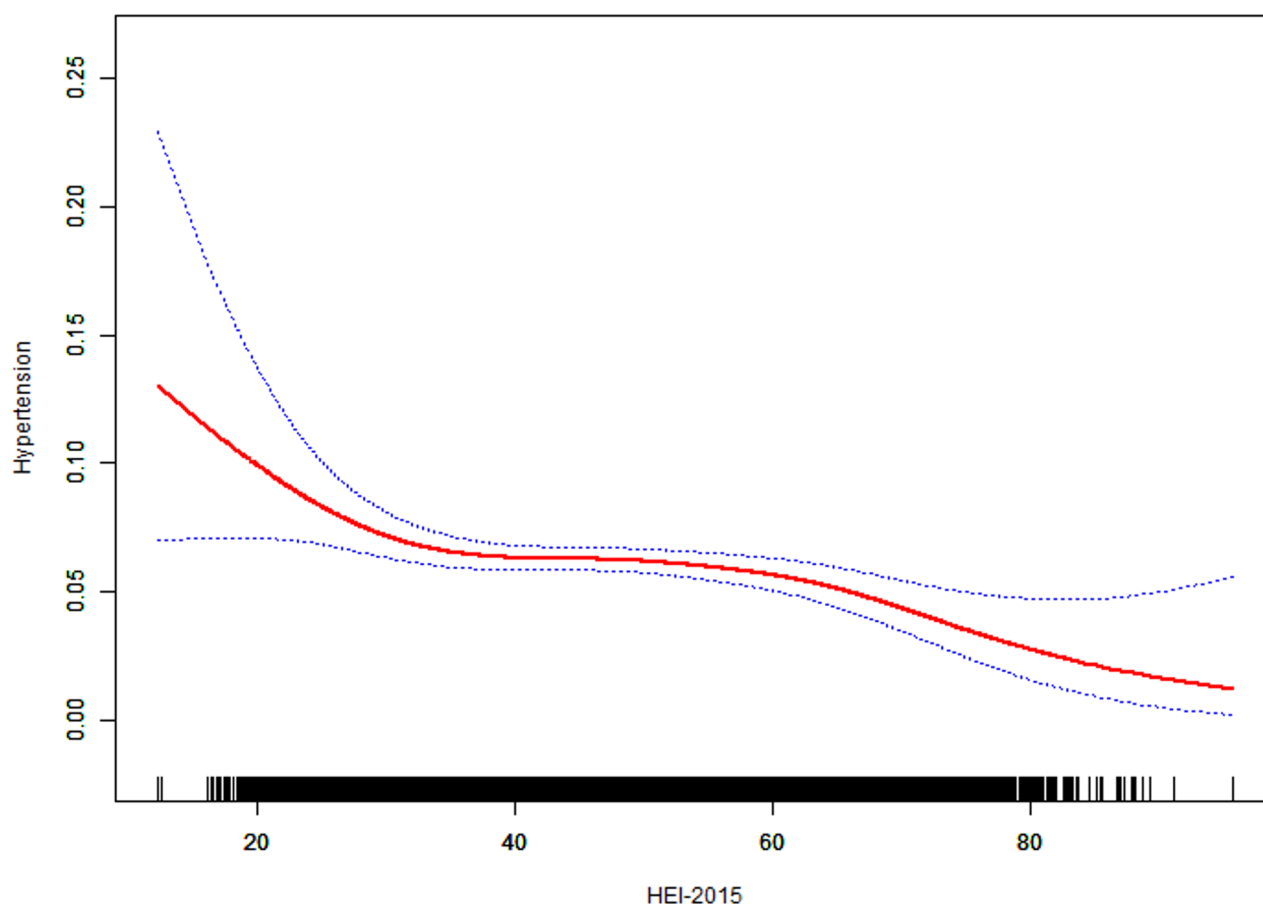


Fig. 2 Associations between the HEI-2015 and hypertension. A threshold, nonlinear association between the HEI-2015 and hypertension was found in a generalized additive model (GAM). Solid red line represents the smooth curve fit between variables. Blue bands represent the 95% of confidence interval from the fit. Adjusted for sex (male or female), age (< 13 years, ≥ 13 years), race/ethnicity (Non-Hispanic White, Non-Hispanic Black, Mexican American, Other Hispanic, other race), Poverty income ratio (Poor, Nearly poor, Middle income, High income, Missing), Education level (Below high school, High school, Above high school), Body Mass Index(kg/m²) Z-score

≥13 years: OR 0.88, $p=0.001$), and major racial/ethnic groups including Non-Hispanic White, Non-Hispanic Black, and Mexican Americans. Protective effects were also significant among nearly poor and middle-income subgroups. Notably, stronger associations were observed in individuals with lower physical activity (<600 MET/week: OR 0.79, $p<0.0001$), higher body weight (>61.3 kg:

OR 0.90, $p=0.008$), and lower lymphocyte counts (OR 0.83, $p=0.001$). Stratification by age, sex, Ethnicity, Poverty income ratio, Education level, revealed no statistically significant interactions (P for interaction > 0.05). Generalized Additive Model curves confirmed consistent inverse relationships across age and sex groups (Fig. 4).

Table 3 Threshold effect analysis of the HEI-2015 and hypertension

Models	HEI-2015 per-unit increase		HEI-2015 per 10unit increase	
	OR (95%CI)	P value	OR (95%CI)	P value
Model I				
One line effect	0.99 (0.98, 0.99)	0.0002	0.90 (0.85, 0.95)	0.0002
Model II				
Turning point(K)	HEI-2015 = 65.13			
HEI-2015 < K	0.99 (0.99, 1.00)	0.0229	0.93 (0.87, 0.99)	0.0231
HEI-2015 ≥ K	0.93 (0.88, 0.98)	0.0086	0.48 (0.28, 0.83)	0.0086
P value for LRT test*	0.012		0.012	
95% CI for turning point	60.15, 66.87			

Data were presented as OR (95% CI), P value; Model I, linear analysis; Model II, non-linear analysis. Adjusted for age (years), sex, Ethnicity, Poverty income ratio, Education level, Body Mass Index(kg/m²) Z-score

CI confidence interval, OR odds ratio, LRT logarithm likelihood ratio test

*P < 0.05 indicates that model II is significantly different from Model I

Table 4 Sensitive analysis of the threshold effect of the HEI-2015 and hypertension

Models	HEI-2015 per-unit increase		HEI-2015 per-unit increase		HEI-2015 per-unit increase		HEI-2015 per-unit increase		HEI-2015 per-unit increase	
	OR (95%CI)	P value	OR (95%CI)	P value	OR (95%CI)	P value	OR (95%CI)	P value	OR (95%CI)	P value
Model I										
One line effect	0.99 (0.98, 0.99)	0.0002	0.99 (0.98, 0.99)	0.0002	0.99 (0.98, 0.99)	0.0002	0.99 (0.98, 0.99)	0.0002	0.99 (0.98, 0.99)	0.0002
Model II										
Turning point(K)	HEI-2015 = 33.05 (Q1)		HEI-2015 = 42.65 (Q2)		HEI-2015 = 50.07 (Q3)		HEI-2015 = 61.96 (Q4)		HEI-2015 = 55.5*	
HEI-2015 < K	0.96 (0.93, 1.00)	0.0332	0.99 (0.97, 1.00)	0.1265	0.99 (0.98, 1.00)	0.1114	0.99 (0.97, 1.00)	0.0327	0.99 (0.99, 1.00)	0.0705
HEI-2015 ≥ K	0.99 (0.99, 1.00)	0.0126	0.99 (0.98, 1.00)	0.0184	0.98 (0.97, 1.01)	0.4107	0.96 (0.92, 0.99)	0.0133	0.98 (0.96, 1.01)	0.1797
P value for LRT test [#]	0.142		0.978		0.409		0.042		0.173	

Sensitivity analysis using pre-specified HEI-2015 cut-offs (the quartiles and HEI categories used in previous work*)

Data were presented as OR (95% CI), P value; Model I, linear analysis; Model II, non-linear analysis. Adjusted for age (years), sex, Ethnicity, Poverty income ratio, Education level, Body Mass Index(kg/m²) Z-score

CI confidence interval, OR odds ratio, LRT logarithm likelihood ratio test

[#]P < 0.05 indicates that model II is significantly different from Model I

Discussion

Our analysis of U.S. children and adolescents observed a significant nonlinear association between overall diet quality, measured by the Healthy Eating Index-2015 (HEI-2015), and hypertension prevalence. Specifically, a threshold effect was identified at an HEI-2015 score of approximately 65. Below this inflection point, the inverse association between HEI-2015 scores and hypertension prevalence was modest. Above this threshold, however, this inverse association was stronger. This indicates that the potential benefits of dietary quality on blood pressure may not be uniform but become more pronounced once diet quality aligns closely with national dietary guidelines. These findings underscore the importance of overall dietary patterns, rather than isolated nutrients, in relation to cardiovascular health indicators in the pediatric population. To our knowledge, this is one of the first

studies to report such a nonlinear, threshold association between diet quality and hypertension in a nationally representative sample of children and adolescents.

Previous studies in adults have consistently shown that higher quality diets are linked to lower incidence of hypertension [23, 36]. Weng et al. analyzed NHANES data in adults and found an inverse association between HEI-2015 and hypertension prevalence [10]. They also noted non-linearity and even a U-shaped association in men (with extremely high HEI men having slightly higher BP, possibly due to other factors). Ducharme-Smith et al. (2021) found that African American adolescents who followed a DASH-style or Alternate Healthy Eating pattern had significantly lower risks of both hypertension and metabolic syndrome compared to those with poorer diets [23]. Previous pediatric studies have typically assessed the diet-hypertension association either as a linear trend

Characteristics	N	OR(95%CI)	P-Value	P for interaction
Age (years)				0.0972
<13	8134	0.890 (0.820, 0.960)	0.0041	
≥13	9394	0.880 (0.820, 0.950)	0.0014	
Sex				0.8177
Male	8766	0.900 (0.840, 0.970)	0.0051	
Female	8762	0.890 (0.820, 0.970)	0.0061	
Ethnicity				0.223
Non-Hispanic White	4818	0.880 (0.780, 0.980)	0.0205	
Non-Hispanic Black	5016	0.860 (0.780, 0.960)	0.005	
Mexican American	4893	0.890 (0.800, 0.990)	0.032	
Other Hispanic	1300	1.130 (0.920, 1.400)	0.2308	
Other Race	1501	0.860 (0.720, 1.040)	0.1202	
Poverty income ratio				0.8067
Poor	4926	0.930 (0.840, 1.030)	0.1519	
Nearly poor	4440	0.870 (0.780, 0.970)	0.0154	
Middle income	4101	0.860 (0.760, 0.960)	0.0101	
High income	2802	0.870 (0.750, 1.010)	0.0631	
Missing	1259	0.940 (0.770, 1.140)	0.5011	
Education level				0.9849
Below high school	11408	0.890 (0.830, 0.960)	0.0026	
High school	4296	0.900 (0.810, 1.000)	0.0428	
Total physical activity (MET/week)				0.0290
<600	4364	0.790 (0.700, 0.880)	<0.0001	
≥600	4191	0.980 (0.870, 1.100)	0.7646	
Missing	8973	0.900 (0.830, 0.970)	0.0042	
Height(cm) group				0.907
105.5 – 149.9	5830	0.890 (0.810, 0.980)	0.0227	
150 – 163.2	5854	0.910 (0.820, 1.020)	0.1077	
163.3 – 200.7	5844	0.880 (0.810, 0.960)	0.0057	
Weight(kg) group				0.9704
17.1 – 45.1	5817	0.910 (0.810, 1.010)	0.0772	
45.2 – 61.2	5854	0.910 (0.820, 1.020)	0.0946	
61.3 – 215.3	5837	0.900 (0.830, 0.970)	0.0078	
BMI Z-score				0.0027
-1.73 – -0.54	5835	0.930 (0.830, 1.050)	0.2288	
-0.54 – -0.18	5836	0.780 (0.700, 0.870)	<0.0001	
-0.18 – -6.78	5837	0.950 (0.880, 1.020)	0.1737	
WBC group				0.0678
2 – 5.8	5155	0.790 (0.710, 0.890)	<0.0001	
5.9 – 7.5	5336	0.870 (0.780, 0.970)	0.0104	
7.6 – 26.3	5373	0.940 (0.860, 1.030)	0.179	
Lymphocyte group				0.3276
0.4 – 1.9	4532	0.830 (0.750, 0.930)	0.0011	
2 – 2.5	5727	0.870 (0.790, 0.970)	0.0118	
2.6 – 6.9	5597	0.930 (0.850, 1.020)	0.1155	

Fig. 3 The association between HEI-2015 and hypertension in subgroups. Data were presented as OR (95% CI), *P* value. HEI-2015, healthy eating index-2015; BMI, Body Mass Index; WBC, white blood cell; OR, odds ratio; CI, confidence interval

or across broad diet quality categories. These methods do not formally test for a change in the slope of the association. Expanding on these findings, our study used piecewise regression and observed a consistent inverse association between HEI-2015 and hypertension across all demographic subgroups analyzed, including both sexes, various age groups, and different racial/ethnic

backgrounds within a nationally representative pediatric sample. A key distinction of our study is the identification of a non-linear, threshold association, with a more pronounced inverse relationship above an HEI-2015 score of approximately 65. Furthermore, in contrast to the U-shaped pattern observed in some adult males [10], our analysis in children and adolescents did not indicate any adverse association at very high HEI-2015 scores. The relationship remained monotonic and beneficial after adjustment for confounders. This difference may be attributable to differences in dietary behavior, metabolism, and blood pressure regulation between children and adults. There are generally less extreme dietary variations or differing physiological responses in younger populations compared to adults.

The mechanisms underlying the inverse non-linear threshold association between diet quality and hypertension remain unclear. Although an HEI score of 65 still represents suboptimal diet quality, it likely represents a meaningful shift toward a dietary pattern characterized by increased consumption of fruits, vegetables, and whole grains, coupled with a reduced intake of added sugars and sodium [16]. The observed inverse association above this threshold may be explained by several factors. First, blood pressure may exhibit nonlinear responses to nutrients, where sufficiently high potassium intake (from fruits/vegetables) combined with sufficiently low sodium intake may trigger significant improvements in vascular endothelial function [37, 38]. The HEI-2015 threshold may mark a critical shift, where combined intake of beneficial nutrients (e.g., potassium, fiber) reaches a level for potential synergistic benefits on vascular endothelial function, alongside meaningful reductions in adverse factors (e.g., sodium, added sugars). This may provide a more plausible biological explanation for the nonlinear

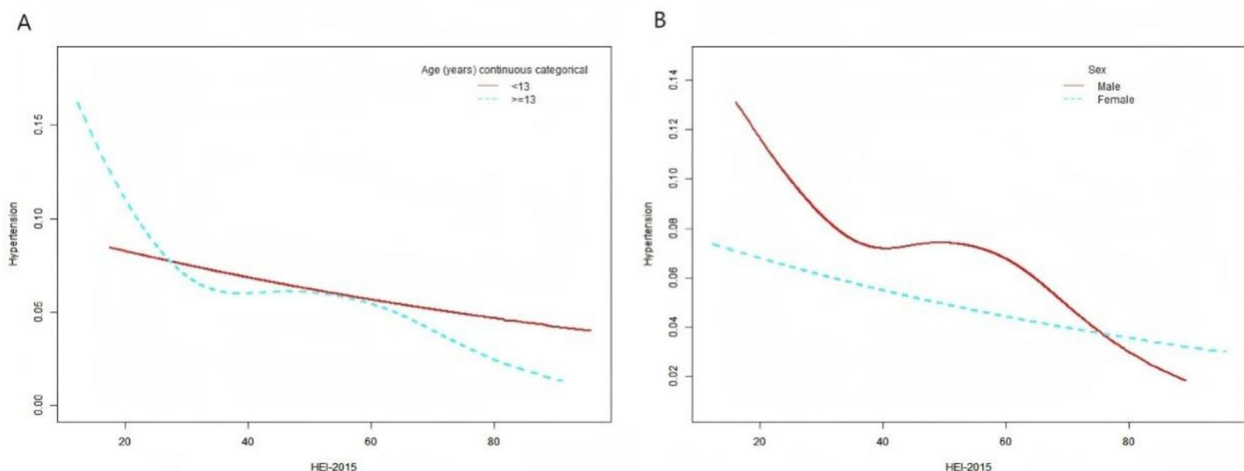


Fig. 4 Generalized Additive Model curve for the association of HEI-2015 with prevalence of hypertension in individuals from NHANES 1999–2020. The association between HEI-2015 and hypertension (A) stratified by sex; (B) stratified by age. HEI-2015, healthy eating index

association. These findings align with DASH diet studies demonstrating that fruit- and vegetable-rich dietary patterns effectively lower blood pressure in both adults and adolescents [39, 40]. While our study was not designed to test specific biological pathways, these explanations remain speculative and are not directly supported by biomarker or nutrient-specific analyses within the study. This threshold may reflect statistical artifact rather than a true physiological inflection point. Second, higher diet quality often coexists with healthier lifestyle behaviors [41]. Thus, the association above the threshold may partially reflect the synergistic benefits of a healthier overall lifestyle, rather than diet alone. These mechanisms remain speculative in the context of our data and warrant investigation in future studies with biomarker measurements. Consequently, this identified threshold may provide a useful reference point for informing future dietary guidelines and lifestyle strategies related to cardiovascular health in the pediatric population.

Consistent with existing study, elevated body mass index (BMI) is a well-established risk factor associated with hypertension [42] and increased weight, and obesity may be major contributors to the hypertension burden [43]. Our study extends this understanding by demonstrating that the inverse association between higher HEI-2015 scores and lower hypertension prevalence remained significant after adjustment for BMI and socioeconomic confounders. This suggests an independent association of overall diet quality with blood pressure outcomes, irrespective of adiposity status [43], which in turn affect hypertension risk. Notably, broader public health improvements, including reductions in sugar-sweetened beverage and sodium consumption, have been speculated to contribute to observed declines in adolescent blood pressure trends [43, 44]. In clinical practice, dietary modification, including caloric restriction, forms a cornerstone of first-line therapy for obesity and its cardiometabolic comorbidities [45]. From a public health perspective, our findings suggest that enhancing diet quality toward higher HEI-2015 scores may be considered in efforts to prevent hypertension in the pediatric population. While the possibility that this threshold reflects statistical artifact rather than a true physiological inflection point, future prospective and interventional studies are required to validate its predictive and causal relationship with hypertension risk in children and adolescents. Specifically, we note that the advanced machine learning algorithms, such as those applied in chronic disease prediction [46], represent a promising future direction for validating and refining dietary threshold models in pediatric populations.

This study has several limitations. First, the cross-sectional nature of our data prevents us from establishing causal relationships. Although we adjusted for multiple

confounders and the biological plausibility supports diet influencing blood pressure, we cannot rule out reverse causation (e.g. children with elevated blood pressure may modify their diets). Second, the composite nature of our hypertension outcome (based on measurement, self-report, and medication) is a limitation, as it may introduce misclassification. Third, dietary intake was assessed using two 24-hour recalls, which, despite being standard in NHANES, remain vulnerable to recall bias, reporting error, and short-term dietary variability that may not accurately reflect habitual intake in children and adolescents. Additionally, the use of HEI-2015 as a composite exposure obscures the contribution of individual dietary components, limiting interpretability of the observed associations. Fourth, we were unable to adjust for important pediatric determinants of blood pressure (e.g., pubertal status, family history, or sleep patterns) due to data limitations, which may affect the causal interpretation of the observed associations. Our adjustment for certain variables, such as BMI-Z score and physical activity are not only potential confounders but may also be mediators on the causal pathway between diet quality and hypertension. By adjusting for them, we may have partially estimated the direct effect of diet that is not mediated through these channels, which could lead to an underestimation of the total effect. Regarding BMI Z-score adjustment, we acknowledge the theoretical concern of overadjustment bias if BMI acts as a mediator. However, our sensitivity analysis (excluding BMI Z-score) yielded nearly identical effect estimates, indicating the core association was robust. This suggests the diet-hypertension link in our sample may operate through pathways largely independent of concurrent BMI, or that BMI's mediating role is minimal.

In contrast, our study has notable strengths. It utilizes a large, nationally representative dataset with rigorous diet and blood pressure measurement protocols, enhancing generalizability to the US youth population. The HEI-2015 effectively captures overall diet quality, and considering non-linear effects adds depth to understanding the diet and blood pressure relationship. Our robust associations remained significant despite adjusting for various confounders, which is timely given emerging preventive cardiology strategies emphasizing the importance of adopting healthy eating patterns from childhood.

Conclusions

In conclusion, this cross-sectional study of U.S. children and adolescents from 1999 to 2020 reveals an inverse nonlinear association between higher adherence to dietary guidelines, measured by HEI-2015 scores, and the prevalence of hypertension in US children and adolescents. Furthermore, our analysis identified a potential inflection point around a score of 65, beyond which the

inverse association was substantially greater. However, the observational design precludes causal inference, and the findings are susceptible to residual confounding, dietary measurement error, and reverse causation. Future prospective and interventional studies are required to validate its predictive and causal relationship with hypertension risk in children and adolescents.

Abbreviations

NHANES	National Health and Nutrition Examination Survey
HEI	Healthy Eating Index
OR	odds ratio
CI	confidence interval
AAP	American Academy of Pediatrics
NCHS	National Center for Health Statistics
USDA	United States Department of Agriculture
NCI	National Cancer Institute
BP	Blood pressure
SBP	Systolic blood pressure
DBP	Diastolic blood pressure
BMI	Body mass index
PIR	Poverty income ratio
ANOVA	One-way analysis of variance
GAM	Generalized additive model
EDU	Education
WBC	White blood cell
LRT	Likelihood Ratio Test
CDC	Centers for Disease Control and Prevention

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s41043-026-01275-0>.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4

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Author contributions

All authors contributed to the study conception and design. Data collection and analysis were performed by DQX and GYQ. BWH, WX and WSF performed statistical analysis. The first draft of the manuscript was written by DQX. LL and ZWR provided guidance in the study design. ZWR contributed to its design and made critical review of the manuscript. All authors commented on previous versions of the manuscript, read and approved the final manuscript.

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Data availability

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Declarations

Ethics approval and consent to participate

The study was approved by the Research Ethics Committee of Jinan Children's Hospital (Children's Hospital Affiliated to Shandong University), and the requirement for informed consent was waived owing to the observation nature of this study. This study was performed in compliance with the Declaration of Helsinki and other relevant regulations.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

Competing interests

The authors declare no competing interests.

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